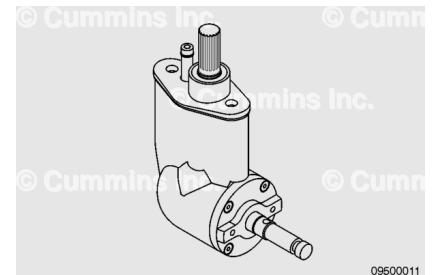


Trainings ▾

General Information

⚠ CAUTION ⚠

Do not operate the thruster without load or out of the water. Damage to the thruster may result due to overspeed. Potential causes of thruster overspeed include but are not limited to: missing or damaged propeller, missing propeller mounting hardware, or damaged coupling. Continued operation of the thruster under these or similar conditions, may result in damage to the thruster.



Note : For original equipment manufacturer (OEM)-supplied component reference equipment manufacturer service information for thruster drive system specifications. For Cummins® thrusters, follow the procedure below.

The drive leg is a lubricated and sealed unit that transfers the rotational energy from the electric motor to the propeller. The gear ratio of the drive leg varies depending on the thruster size.

When installed in a metal thrust tunnel, it is important that the drive leg be completely electrically insulated from the tunnel to prevent galvanic corrosion to the thrust tunnel. The gaskets supplied are electrically insulated. The capscrews and shaft **must** be fitted with insulation material, such as nylon bushings.

Fiberglass tunnels do **not** need to be electrically insulated from the drive leg.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Always remove the vessel from the water before removing the drive leg, mounting flange, or thruster tunnel. Failure to remove the vessel from the water prior to removing these parts will cause sea water to enter the vessel, possibly resulting in property damage.

- Remove the vessel from the water prior to removing the drive leg to prevent sea water from entering the vessel.
- Turn the thruster power disconnect switch to the disconnect position. Refer to Procedure 013-079 in Section 13. (</qs3/pubsys2/xml/en/procedures/326/326-013-079.html>)
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the propeller. Refer to Procedure 009-062 in Section 9. (</qs3/pubsys2/xml/en/procedures/326/326-009-062.html>)
- Remove the direct current motor. Refer to Procedure 013-077 in Section 13. (</qs3/pubsys2/xml/en/procedures/326/326-013-077.html>)

Remove

Remove the two capscrews that secure the mounting flange to the thruster tunnel.

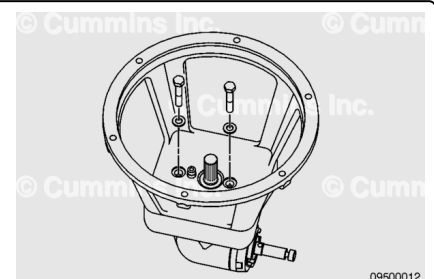
Remove the drive leg. Remove and discard the gasket(s). Record the number of gaskets between the drive leg and the thruster tunnel. The same number of gaskets will be used for installation to make sure there is proper alignment of the drive leg and propeller.

Note : The plastic spacer mounted on the drive leg that mates to the thruster tunnel should not be removed from the drive leg.

Remove the mounting flange.

Discard the gasket between the mounting flange and the thruster tunnel.

Note : The two capscrews secure both the mounting flange and drive leg. Removing them will cause the drive leg to dismount from the tunnel.



Clean and Inspect for Reuse

⚠ WARNING ⚠

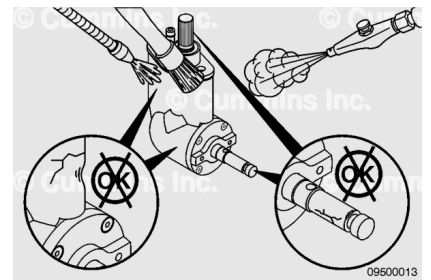
To reduce the possibility of personal injury, wear goggles and protective clothing.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

The propeller is made of acetyl plastic which is not compatible with acids. Exposure to acid cleaners (example: muriatic acid or hydrochloric acid such as MaryKate On and Off) will cause the immediate loss of strength of the propeller and cause cracking.



Clean the drive leg with a solvent and dry thoroughly.

Inspect the drive leg for leaks, bent shaft, or other damage. If any damage is found, replace the part.

Install

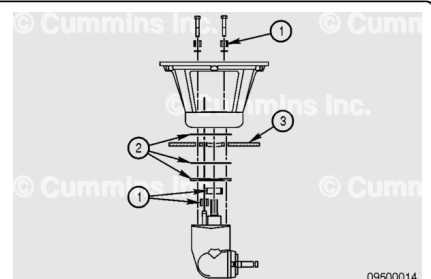
Apply a sealant such as silicone or polyurethane to both sides of the new gaskets.

Install a new gasket(s) onto the drive leg. Use the same number of gaskets found during removal.

Install the drive leg.

Install a new gasket onto the bottom of the mounting flange.

Grease the hole of the mounting flange with outboard gear grease and install the flange.



Grease the threads with outboard gear grease and install the two capscrews that mount the drive leg and mounting flange to the thruster tunnel.

Thruster sizes 95 kg [209 lb] and below use M8 capscrews.

Torque Value: 13 n•m [115 in-lb]

Thruster sizes 125 kg [276 lb], 160 kg [353 lb], and 220kg [485 lb] use M12 capscrews.

Torque Value: 33 n•m [24 ft-lb]

Inspect the gap between the propeller and thruster tunnel to check for a uniform gap.

Check that the propeller spins freely without rubbing. The gap between the propeller and tunnel should always be between 2.4 mm [0.09 in] and 5.6 mm [0.22 in]. The gap should never exceed 6 mm [0.24 in] as the thruster will no longer produce adequate thrust. A smaller gap is better and produces more thrust. If the propeller does **not** fit inside the tunnel, the propeller blades can be shaved down to fit the tunnel. The blades **must** be shaved down evenly.

Note : Insulated bushings are **only** needed for metal thrust tunnels to prevent electrolysis corrosion.

Note : The gasket(s) between the drive leg and thruster tunnel are each 2 mm [0.07 in] thickness. The gasket between the mounting flange and thruster tunnel is 1 mm [.04 in] thickness.

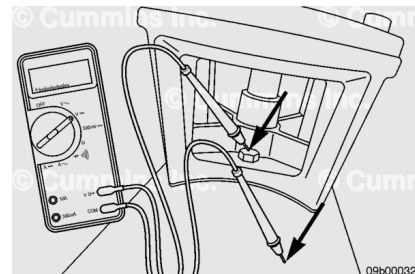
Test

Isolation test for metal hulls:

Note : The boat **must** be out of the water for the isolation test results to be accurate.

Once the thruster is installed in a metal boat, isolation should be verified.

Make sure that there is **no** electrical wiring attached to the thruster (either power or control harnesses).



Use a multimeter to measure resistance between the metal hull (on a clean, unpainted surface) and the thruster (at the mounting screws of the mounting flange).

There should be greater than 1 megohm resistance between these points. If not, re-check the mounting of the thruster.

Note : In order to prevent possible galvanic corrosion, the thruster must be mounted using the appropriate bushings and gaskets to create complete electrical isolation from a vessel with a metal hull. Contact the vessel original equipment manufacturer (OEM) for isolation kit information.

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the direct current motor. Refer to Procedure 013-077 in Section 13 (</qs3/pubsys2/xml/en/procedures/326/326-013-077.html>).
- Install the propeller. Refer to Procedure 009-062 in Section 9. (</qs3/pubsys2/xml/en/procedures/326/326-009-062.html>)
- Install the propeller. Refer to Procedure 009-062 in Section 4. (</qs3/pubsys2/xml/en/procedures/326/326-009-062.html>)
- Connect the batteries. See equipment manufacturer service information.
- Turn the thruster power disconnect switch to the connect position. Refer to Procedure 013-079 in Section 13. (</qs3/pubsys2/xml/en/procedures/326/326-013-079.html>)
- Launch the vessel back into the water. Check for leaks.